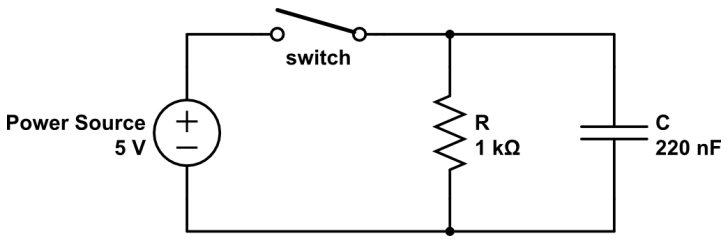
Circuit Lab Answer Key

1. 267 Ω *(3 points)*
2. greater than *(1)*
3. decrease *(1)*
4. 2.25 V *(1)*
5. $I_{R6} = \frac{1}{2} I_{R3}$ *(1)*
6. 50.6 mW *(2)*
7. 1A *(2)*
8. $I_{R1} = \pm(I_{R3} - I_{R2})$ *(2)*
9. 3 V *(2)*
10. 14 V *(2)*
11. 0 A *(1)*
12. 220 µs *(2)*
13. 220 nF *(2)*
14. 5 mA *(2)*
15. Write answer below *(5)*
16. 23.5 µA *(1)*
17. 10.1 nF *(1)*
18. 57.3 µA *(2)*
19. slower *(1)*
20. $V_{C2}(t) = 2.84 + 3.04e^{-917t}$ *(5)*
21. b *(2)*
22. d *(2)*
23. a *(2)*
24. e *(2)*
25. c *(2)*
26. diode *(1)*
27. ∞ Ω (infinite) *(2)*
28. 0 Ω *(2)*
29. Ampere *(2)*
30. Volt *(2)*
31. tolerance *(2)*

15. Circuit diagram:



32. What is the purpose of a Wheatstone Bridge? *(3 points)*

The Wheatstone Bridge allows measurement of an unknown resistance to a very high degree of precision. It uses a diamond shape, where one of the sides is an unknown resistance and the rest are known. The resistance can be found by measuring the current flow through the middle of the diamond when the sides are connected.

33. Define resistivity. *(2 points)*

Resistivity is a material property. It is the inverse of conduction and measures a material's resistance to the flow of electrons per unit length through a certain cross sectional area. Its units are ohm-meters.

34. Define drift velocity. *(2 points)*

Drift velocity is the overall speed and direction that electrons travel through a material, not including any components of movement perpendicular to the direction of motion. Current (measured in amperes), implicitly uses drift velocity when talked about in terms of the speed of electron flow.

Note: Answers for the 2 Lab Stations are omitted. Answers were based upon the actual voltage of the 9V batteries used. Be aware that the circuits at both stations were built on a breadboard.

Virginia - Repaired practice edition - Answer key

Division C | Circuit Lab | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

repair-case. (6 points) Authored written substitute for unavailable source material: Solve this case and explain your reasoning: Two point charges keep their values while separation triples. Predict force.

reasoning: Use F proportional to 1/r^2.
Replacing r with 3r multiplies the denominator by nine.
The new magnitude is F/9; attraction or repulsion is unchanged. (3 points).

conclusion: Direction depends on signs, not distance alone. (3 points).